# PCEOT402
# ELECTRICAL MACHINES

Module 3

Synchronous Machines

# Three phase Synchronous Machine

- Principle of Operation of 3-phase synchronous generator – classification -
- constructional features - EMF equation – coil-span factor and distribution factor (sinusoidal flux distribution only) – numerical problems Synchronous generator on no-load – open circuit characteristics –
- synchronous generator on load – armature reaction – effect of armature reaction
- Equivalent circuit - phasor diagram – voltage regulation –
- predetermination of voltage regulation by emf and mmfmethod –
- Parallel operation - synchronous generator on infinite bus-bar – conditions – methods of synchronisation – dark lamp method – bright lamp method (2 hours)
- Synchronous motor – rotating magnetic field (no derivation) - principle of operation – starting methods

- Principle of Operation of 3-phase synchronous generator

# AC Generator

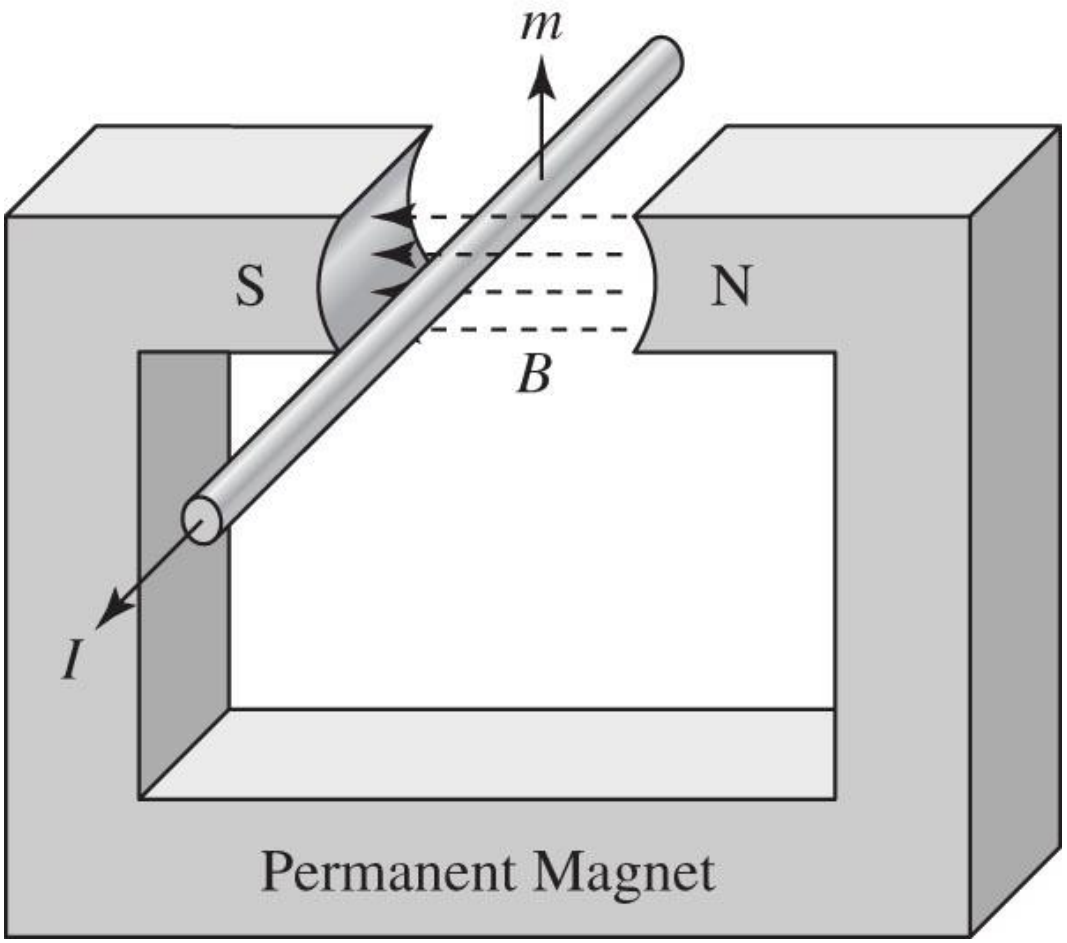
- Faraday's law dictates that if a coil of *N* turns experiences a change in magnetic flux, then the induced voltage *V* is given by

$$V = N\frac{\mathrm{d}\Phi}{\mathrm{d}t}$$

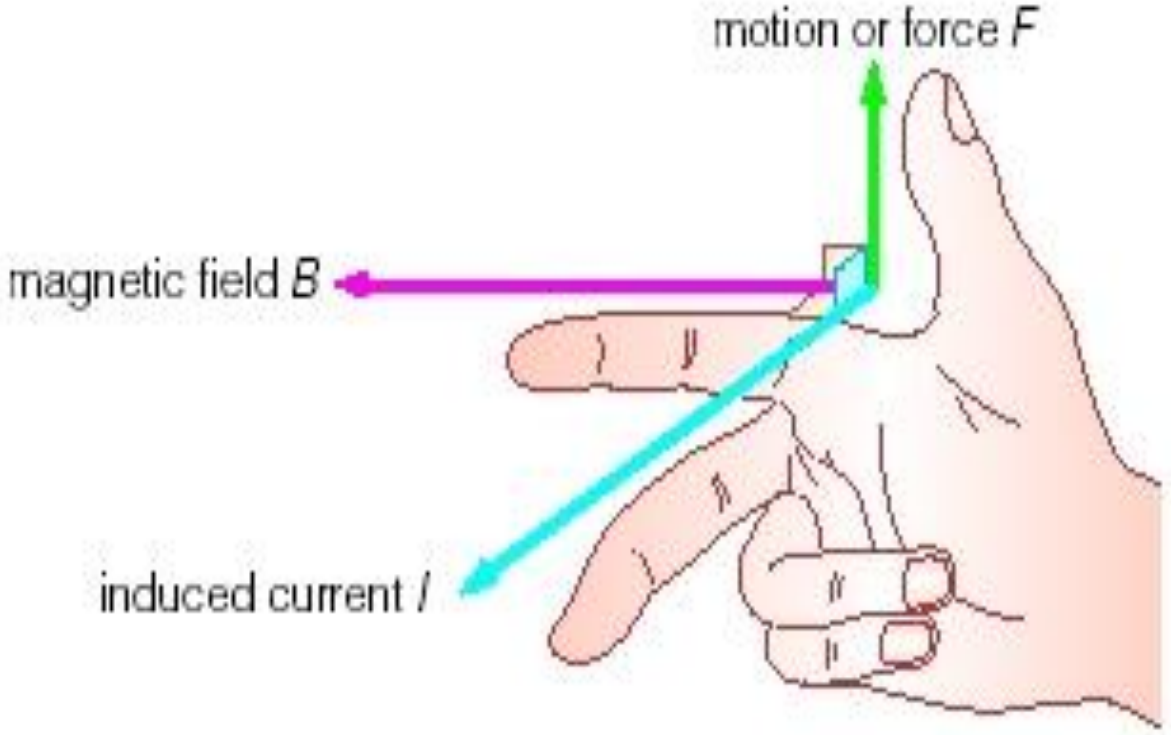
- If a coil of area *A* rotates with respect to a field *B*, and if at a particular time it is at an angle $\theta$ to the field, then the flux linking the coil is $BA\cos\theta$, and the rate of change of flux is given by

$$\frac{d\Phi}{dt} = BA\frac{\mathrm{d}(\sin\theta)}{\mathrm{d}t} = \frac{\mathrm{d}\theta}{\mathrm{d}t}\cos\theta = \omega\cos\theta$$

# Magnetic induction in a wire moving in a field.



# Fleming's Right hand rule

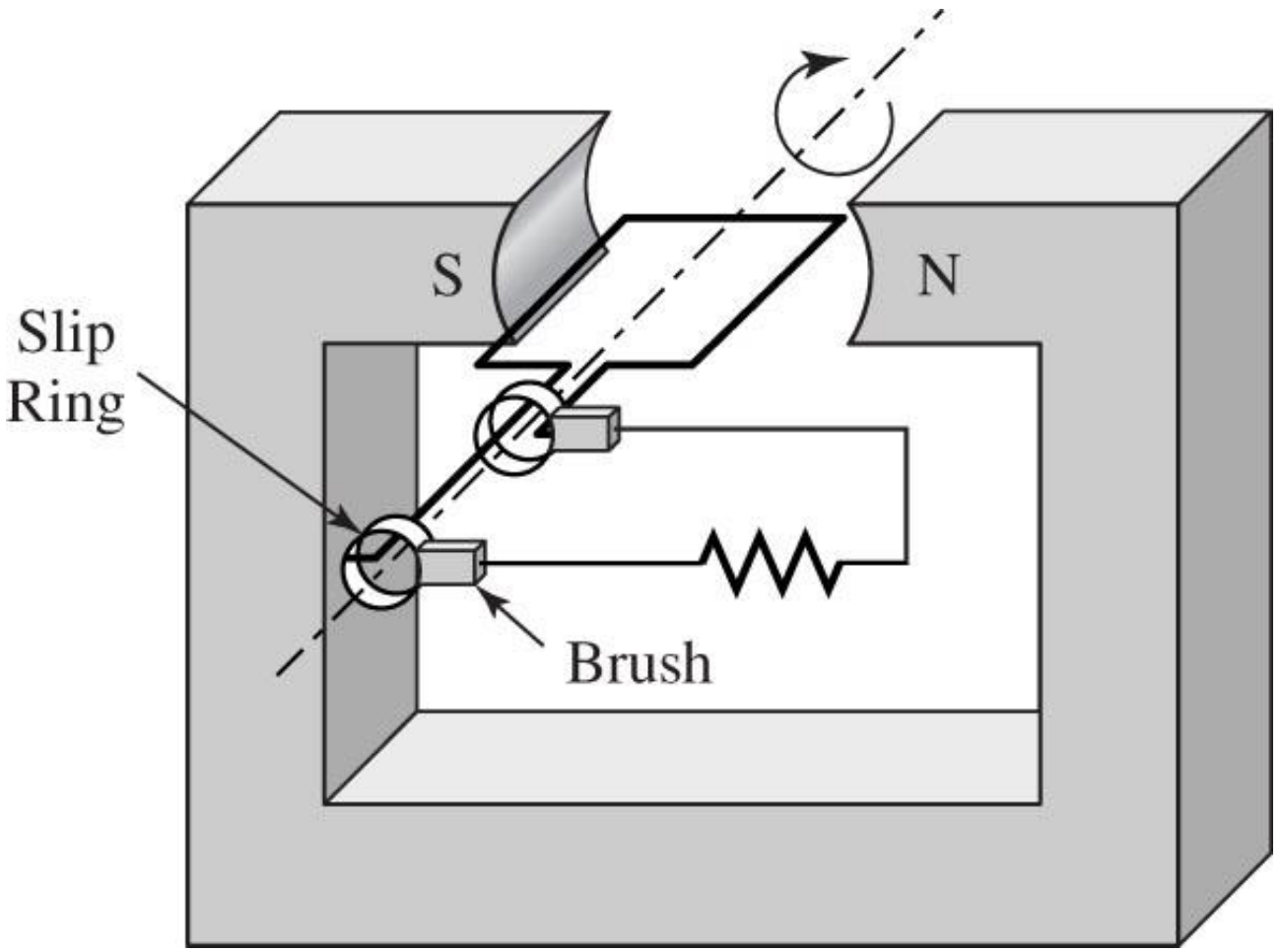


# Fleming's Right hand rule

- Used to determine the direction of emf induced in a conductor
- The middle finger , the fore finger and thumb of the left hand are kept at right angles to one another.
  - The fore finger represent the direction of magnetic field
  - The thumb represent the direction of motion of the conductor
  - The middle finger will indicate the direction of the inducted emf .

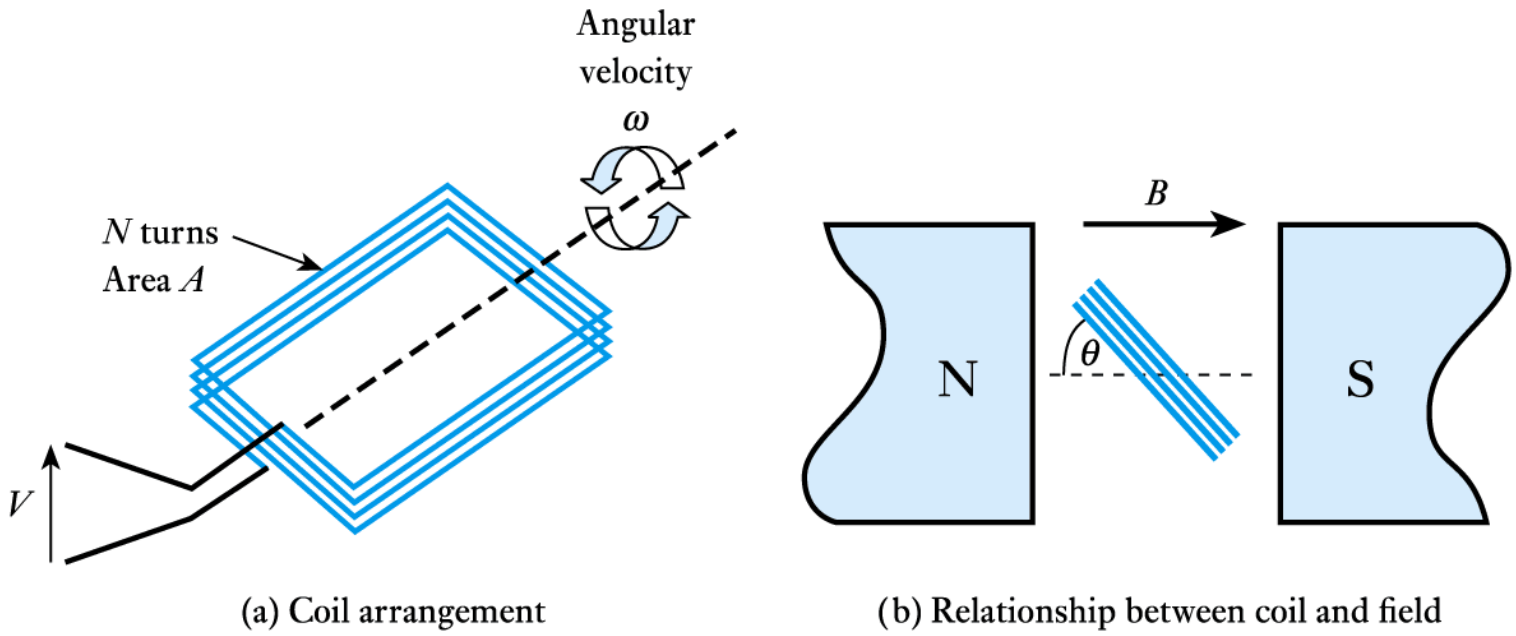
This rule is used in DC Generators

# AC generator with slip rings and brushes.



- Thus for the arrangement shown below

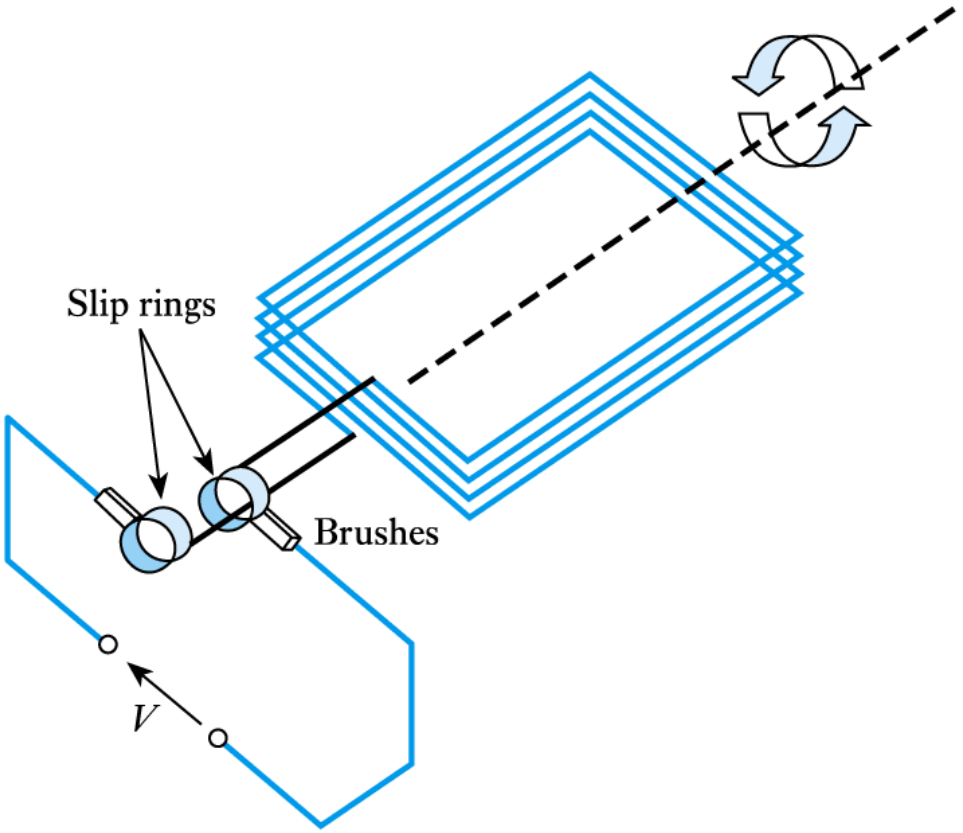
$$V = N\frac{\mathrm{d}\Phi}{\mathrm{d}t} = NBA\frac{\mathrm{d}(\sin\theta)}{\mathrm{d}t} = NBA\omega\cos\theta$$



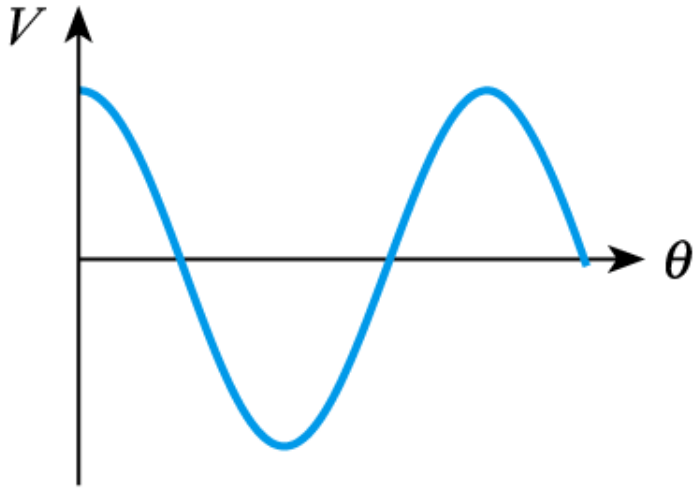

(a) Coil arrangement

(b) Relationship between coil and field

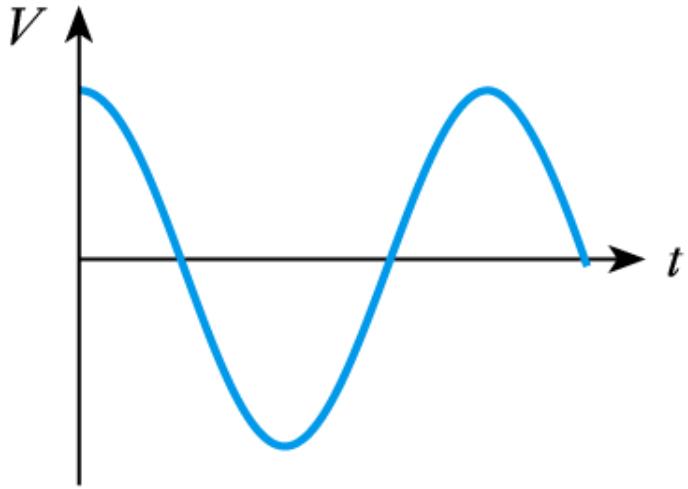
- Wires connected to the rotating coil would get twisted
- Therefore we use circular **slip rings** with sliding contacts called **brushes**



- Therefore this arrangement produces a sinusoidal output as shown below




(a)




(b)

# SYNCHRONOUS GENERATOR CONSTRUCTION

- **rotor carry the "field windings" , while "armature windings" (or "stator windings") carry the main voltages of machine**

  in synchronous generator, rotor winding energized by dc source to develop rotor magnetic field

- rotor is turned by a prime mover, producing a rotating magnetic field which induce 3 phase voltages in stator windings

**therefore:**

- rotor windings - field windings
- stator windings - armature windings

# 3-phase synchronous generator – classification

**Based on Prime Mover**

- **Hydro Generators** – Driven by water turbines
- **Turbo Generators** – Driven by steam turbines
- **Diesel Engine Generators** – Driven by diesel engines
- **Gas Turbine Generators** – Driven by gas turbines

**Based on No. of Phase**

- Single phase
- Three phase

**Based on type of roatating**

- Rotating Armature
- Rotating Field

**Based on Rotor Construction**

- Salient Pole machine
- Non Salient Pole Machine

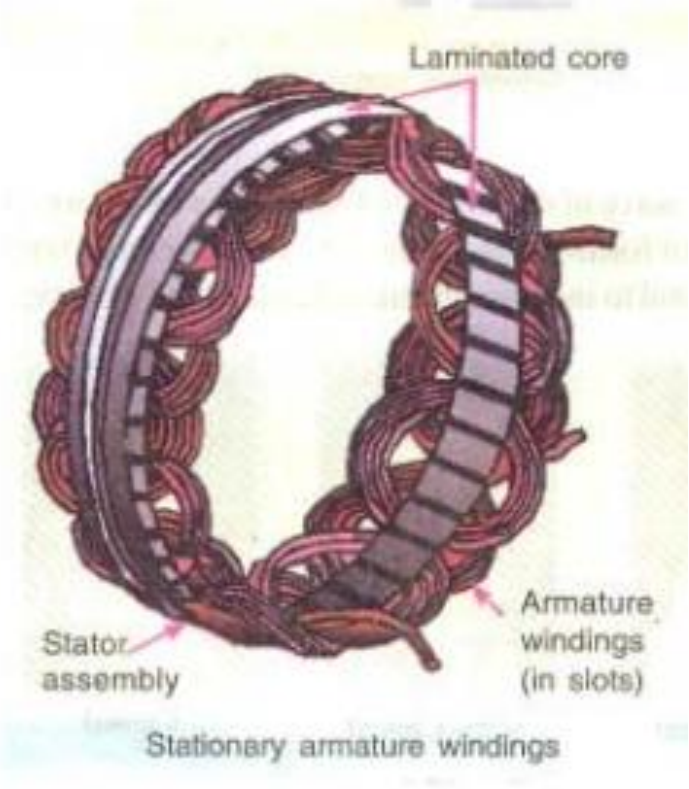
# Advantages of rotating field and stationary armature type

- High voltage generation
- Better insulation
- Rotor weight – less
- Current collection - easy
- Lesser number of slip rings

# Alternator - Construction

Standard construction consists of :

- Armature windings on stationery stator
- Field poles on rotor.
- The field windings are excited by DC supply from the exciter and given through the slip rings

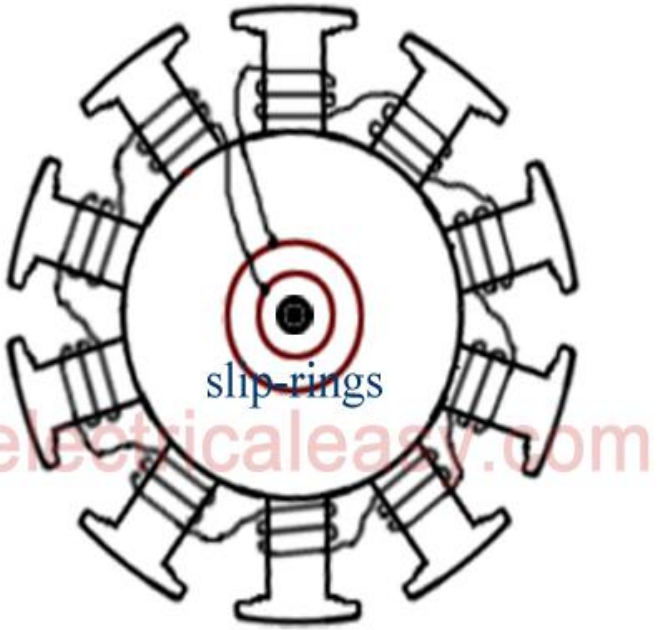


# Types of alternators based on rotor construction

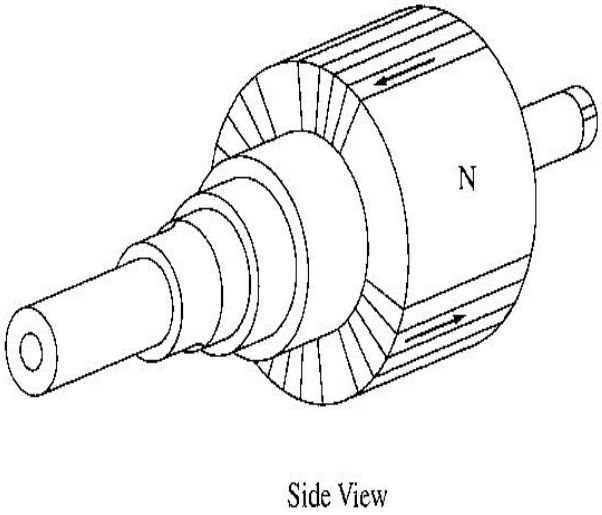
- Salient pole alternator
- Non salient pole or Turbo or Cylindrical rotor alternator

# Rotor

Salient pole alternator

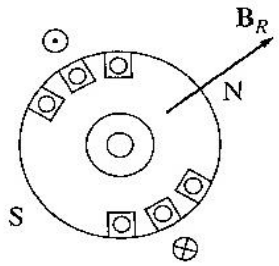


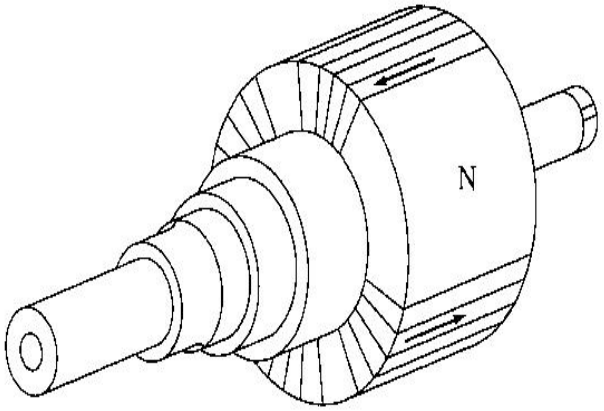

Non salient pole or Turbo or cylindrical rotor alternator



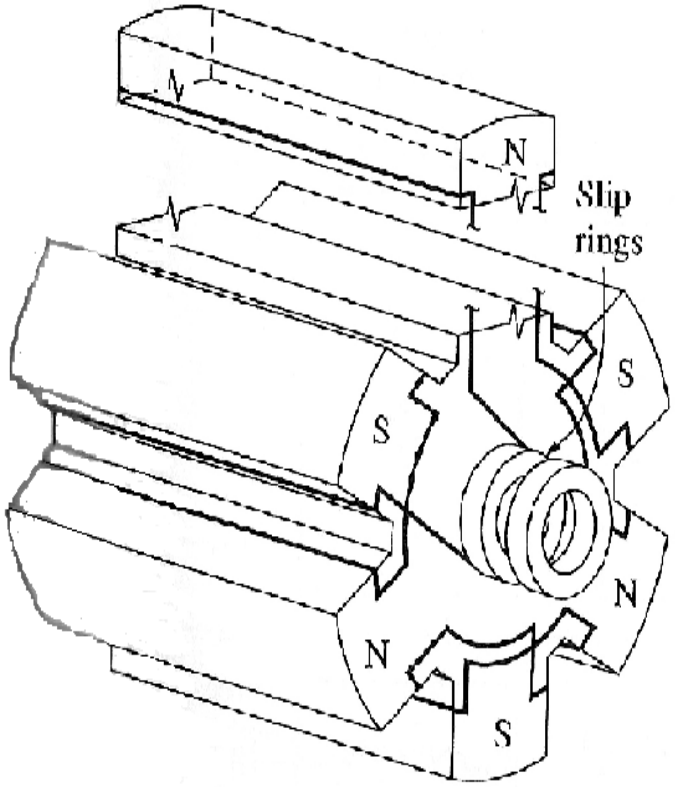
- Rotor of synchronous machine can be

**Nonsalient:** 2 pole rotor **Salient:** six-pole rotor




Side View



# Salient pole type alternator

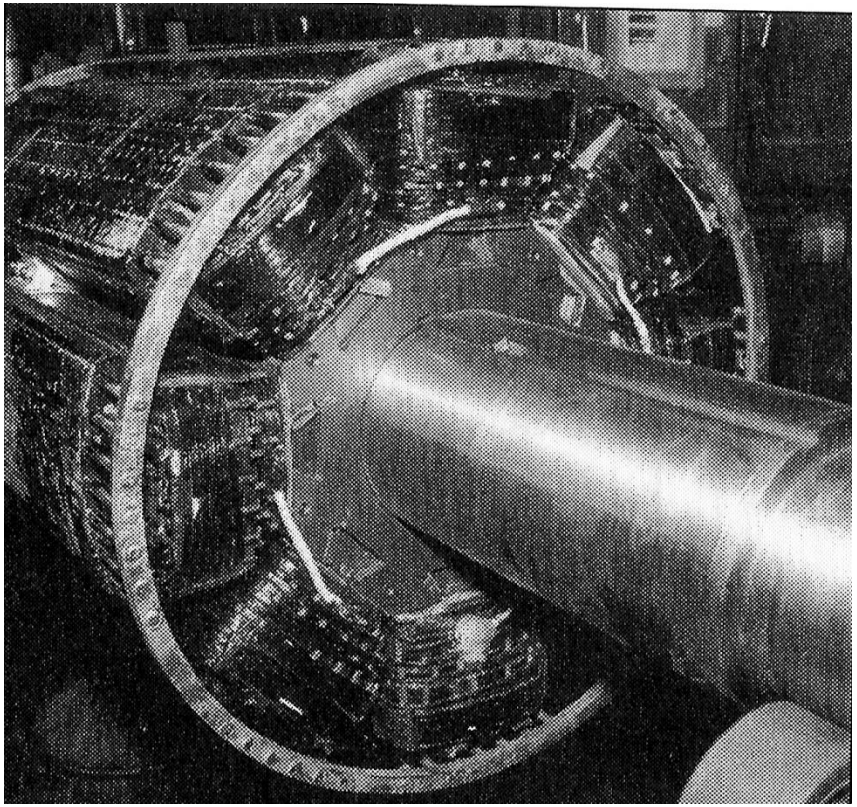
- It consists of cast iron frame which supports the silicon steel laminated ( bundling of thin sheets) armature core .
- The core has slots on its inner periphery for housing the conductor .
- Silicon steel material reduces the hysteresis loss and the lamination structure reduces the eddy current loss.
- Suitable for low and medium speed alternator ( Ex Hydro alternator )

# Advantages of a smooth cylindrical type rotor

- Less windage loss
- No need of damper winding
- Good balancing of the rotor
- Sound less running
- Good cooling to the field because it is a distributed one.

# SYNCHRONOUS GENERATOR CONSTRUCTION

- Photograph of a salient 8-pole synchronous machine rotor



# SYNCHRONOUS GENERATOR CONSTRUCTION

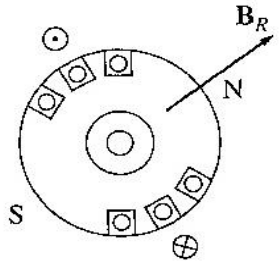
- **Rotor** experience varying magnetic fields, therefore is constructed of thin laminations to reduce eddy current losses
- To supply the rotor winding while it is rotating, special arrangement employed to connect its terminal to dc supply

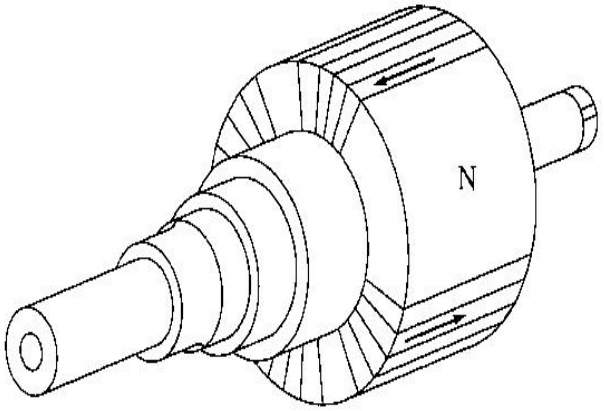
1. supply dc power from an external dc source to rotor by means of slip rings
2. supply dc power from a special dc power source mounted on shaft of rotor

# SYNCHRONOUS GENERATOR CONSTRUCTION

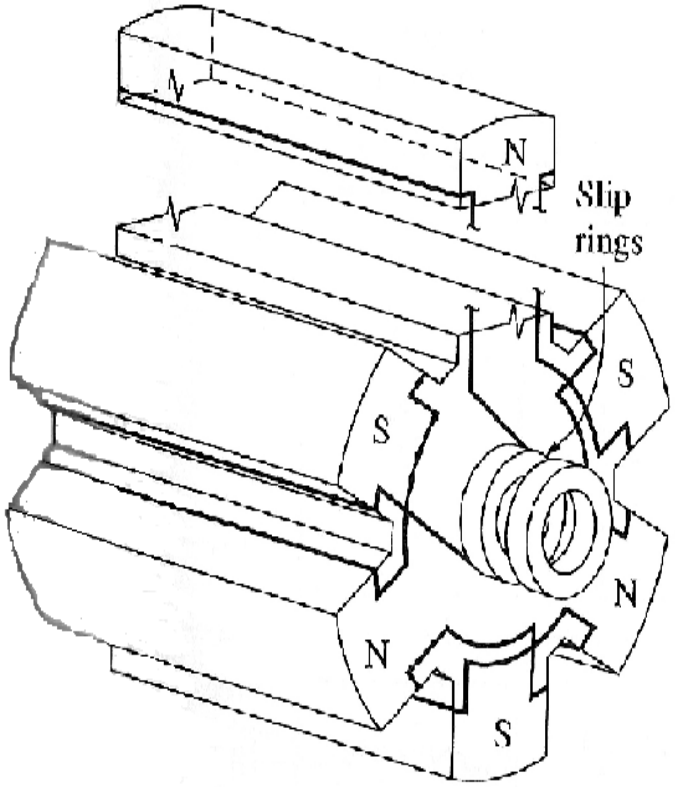
- Rotor of synchronous machine can be

**Nonsalient:** 2 pole rotor **Salient:** six-pole rotor



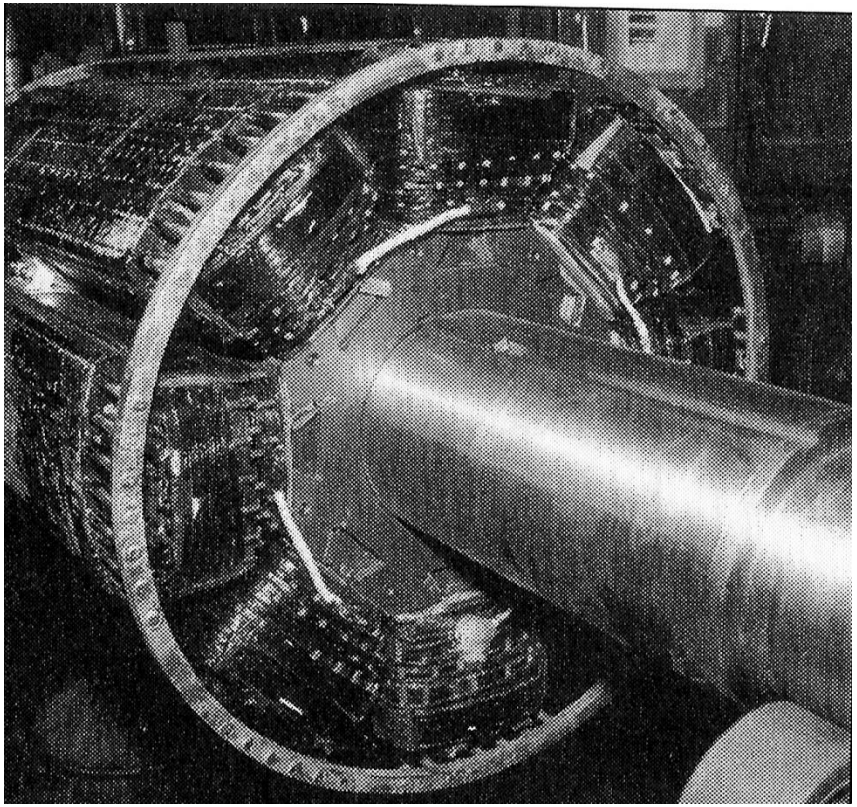

Side View



| | Salient Pole Rotor | Non-Salient Pole (Cylindrical Rotor) |
|---|---|---|
| **Rotor Construction** | Projecting poles mounted on rotor | Smooth solid cylindrical rotor |
| **Air Gap** | Non-uniform air gap | Uniform air gap |
| **Shape** | Large diameter, short axial length | Small diameter, long axial length |
| **Speed of Operation** | Low speed (100–600 rpm) | High speed (1500–3000 rpm) |
| **Number of Poles** | Large number of poles (6–60 poles) | Usually 2 or 4 poles |
| **Mechanical Strength** | Lower mechanical strength | Very high mechanical strength |
| **Applications** | Hydroelectric power plants | Thermal & nuclear power plants |
| **Cooling** | Natural or air cooling | Hydrogen or forced cooling (in turbo generators) |

# SYNCHRONOUS GENERATOR CONSTRUCTION

- salient 8-pole synchronous machine rotor



# SYNCHRONOUS GENERATOR CONSTRUCTION

- **Rotor** experience varying magnetic fields, therefore is constructed of thin laminations to reduce eddy current losses
- To supply the rotor winding while it is rotating, special arrangement employed to connect its terminal to dc supply

1. supply dc power from an external dc source to rotor by means of slip rings
2. supply dc power from a special dc power source mounted on shaft of rotor

paper.

Fig. 37.6

When the conductor is in the interpolar gap, as at *A* in Fig. 37.7, it has minimum e.m.f. induced in it, because flux density is minimum there. Again, when it is at the centre of a *S*-pole, it has maximum e.m.f. induced in it, because flux density at *B* is maximum. But the direction of the e.m.f. when conductor is over a *N*-pole is opposite to that when it is over a *S*-pole.

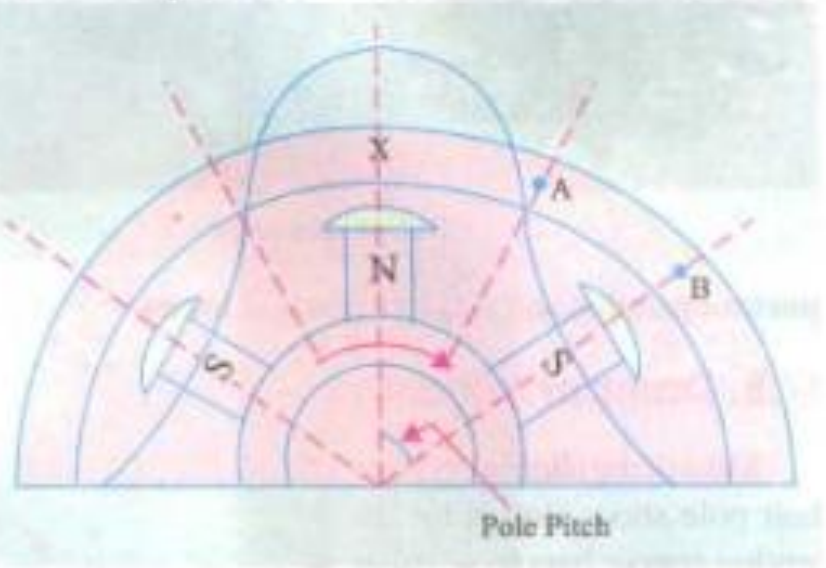



Fig. 37.7

Obviously, one cycle of e.m.f. is induced in a conductor when one pair of poles passes over it. In other words, the e.m.f. in an armature conductor goes through one cycle in angular distance equal to twice the pole-pitch, as shown in Fig. 37.7.

Let $P$ = total number of magnetic poles

$N$ = rotative speed of the rotor in r.p.m.

$f$ = frequency of generated e.m.f. in Hz.

Since one cycle of e.m.f. is produced when a pair of poles passes past a conductor, the number of cycles of e.m.f. produced in one revolution of the rotor is equal to the number of pair of poles.

$\therefore$ No. of cycles/revolution = $P/2$ and No. of revolutions/second = $N/60$

$$\therefore \quad \text{frequency} = \frac{P}{2} \times \frac{N}{60} = \frac{PN}{120} \text{ Hz}$$

or
$$f = \frac{PN}{120} \text{ Hz}$$

$N$ is known as the synchronous speed, because it is the speed at which an alternator must run, in

# SYNCHRONOUS GENERATOR

## Speed of rotation of synchronous generator

- synchronous generators are *synchronous, during their operation*

means: electrical frequency is synchronized with mechanical speed of rotor

- Relation between electrical frequency of stator and mechanical speed of rotor as shown before:
- $f = Ns\ p / 120$

f : electrical frequency in Hz

Ns: speed of rotor in r/min

p: number of poles

# Winding factor

The product of pitch factor $K_p$ and distribution factor $K_d$ is referred to the winding factor

$$i.e.\ K_w = K_p K_d.$$

The pitch factor and distri...

# Pitch factor

**3. 6.1. Pitch Factor.** In a short pitch winding the induced emfs in the two sides of the coil are not in phase, as discussed in Art 3.5, hence their resultant given by the phasor sum is always less than their arithmetic sum. The ratio of phasor sum of induced emfs per coil to the arithmetic sum of induced emfs per coil is known as *coil pitch factor, coil-span factor* or simply *pitch factor* ($K_c$ or $K_p$). It is always less than unity.

Let the coil have a pitch short by an angle $\alpha$ electrical space degrees from full pitch and induced emf in each coil side be E.

Now if the coil would have been full-pitched, then total induced emf in the coil would have been 2 E.

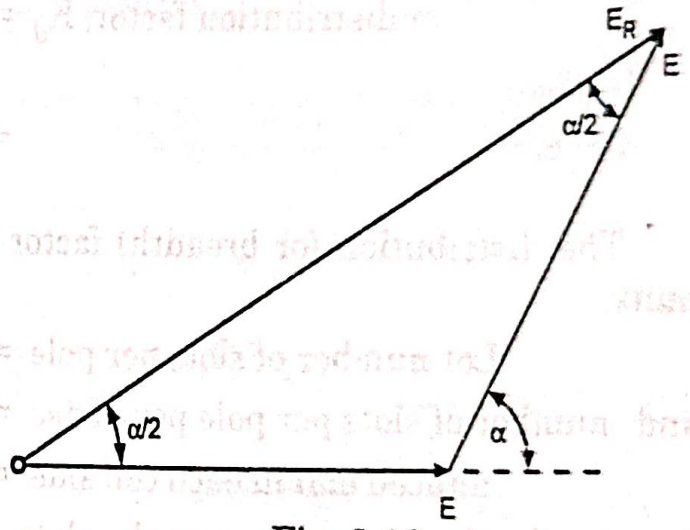



*Fig. 3.16*

If the coil is short-pitched by $\alpha$ electrical space degrees the resultant induced emf, $E_R$ in the coil is phasor sum of two voltages $\alpha^\circ$ (electrical) apart

$$i.e.\ E_R = 2\,E\cos\frac{\alpha}{2}$$

$$\text{Pitch factor, } K_p = \frac{\text{Phasor sum of coil side emfs}}{\text{Arithmetic sum of coil side emfs}} = \frac{2\,E\cos\frac{\alpha}{2}}{2\,E} = \cos\frac{\alpha}{2} \qquad ...(3.3)$$

# Distribution factor

*The ratio of the phasor sum of the emfs induced in all the coils distributed in a number of slots under one pole to the arithmetic sum of the emfs induced (or to the resultant of emfs induced in all coils concentrated in one slot under one pole) is known as breadth factor ($K_b$) or distribution factor ($K_d$)*

$$\text{or distribution factor, } K_d = \frac{\text{EMF induced in distributed winding}}{\text{EMF induced if the winding would have been concentrated}}$$

$$= \frac{\text{Phasor sum of component emfs}}{\text{Arithmetic sum of component emfs}}$$

The distribution (or breadth) factor is always less than unity.

Let number of slots per pole $= n$

and number of slots per pole per phase $= m$

Induced emf in each coil side $= E_c$

Angular displacement between the slots,

$$\beta = \frac{180^\circ}{n}$$

The emfs induced in different coils of one phase under one pole are represented by side AC, CD, DE, EF, ...which are equal in magnitude (say each equal to E) and differ in phase (say by β°) from each other.

If bisectors are drawn on AC, CD, DE, EF... they would meet at common point (say at O). The point O would be the circum centre of the circle having AC, CD, DE, EF...as the chords and representing the emfs induced in the coils in different slots.

$$\text{EMF induced in each coil side, } E = AC = 2\,OA \sin\frac{\beta}{2}$$

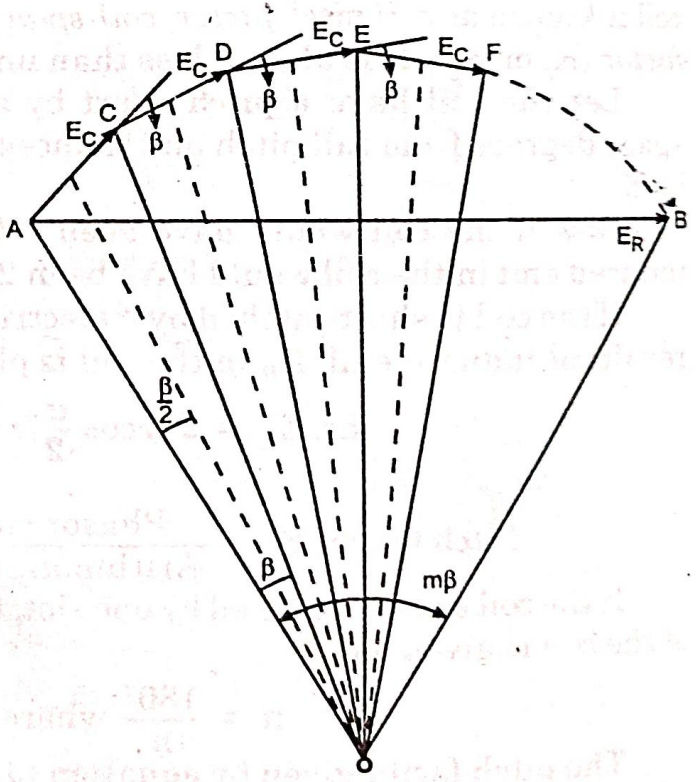



*Fig. 3.17*

$$\text{Arithmetic sum} = m \times 2 \times \text{OA} \sin \frac{\beta}{2}$$

The resultant emf induced in one polar group of one phase would be the phasor sum as represented by the phasor AB shown in fig 3.17.

$$\therefore \text{The resultant emf, } E_R = \text{AB} = 2 \times \text{OA} \sin \frac{\text{AOB}}{2} = 2 \times \text{OA} \sin \frac{m\beta}{2}$$

$$\text{and distribution factor, } K_d = \frac{\text{Phasor sum of components emfs}}{\text{Arithmetic sum of components emfs}}$$

$$= \frac{2 \times \text{OA} \sin \frac{m\beta}{2}}{m \times 2 \times \text{OA} \sin \frac{\beta}{2}} = \frac{\sin \frac{m\beta}{2}}{m \sin \frac{\beta}{2}} \qquad \ldots(3.5)$$

$m\beta$ is also known as the *phase spread* and is spread in electrical degrees or radians.

The distribution factor $K_d$ given by equation (3.5) is for the fundamental component of emf. If the flux distribution contains space-harmonics, the slot angular pitch $\beta$ on the fundamental scale, would become $r\beta$ for the *r th* harmonic component and thus the distribution factor for the *r th* harmonic would be

$$K_{dr} = \frac{\sin \frac{rm\beta}{2}}{m \sin \frac{r\beta}{2}} \qquad \ldots(3.6)$$

# EMF Equation

Let $Z_p$ = Number of conductors in series per phase

P = Number of poles

$\phi$ = Useful flux per pole in webers

N = Rotational speed in rpm

and $f$ = Frequency in herz and is equal to $\frac{NP}{120}$

The flux cut by any conductor while passing from the centre of one inter polar gap to the centre of the next is $\phi$ webers and since during the movement, the emf wave completes half cycle *i.e.* the time taken is $\frac{1}{2f}$ seconds, therefore, the average rate of cutting the flux,

$$\frac{d\phi}{dt} = \frac{\phi}{1/2f} = 2f\phi \text{ Wb/s}$$

Hence average emf induced in each conductor = $2f\phi$ volts

Average emf per phase, $E_{av}$/phase = Number of conductors in series/phase arranged in one slot/pole × average emf induced per conductor

$= Z_p \times 2f\phi = 2T \times 2f\phi = 4\phi f T$ volts $\quad \because Z_p = 2T$ (turns)

For distributed winding the average value of emf per phase will be $K_d$ times the above value

*i.e.* $E_{av}$ / phase $= 4K_d \phi f T$ volts

For short pitch winding the true average value of emf per phase will be $K_p$ time above value

$E_{av}$/phase $= 4K_d K_p \phi f T$ volts

and $E_{rms}$/phase = Form factor, $K_f \times E_{av}$/phase $= 4K_f K_d K_p \phi f T$ volts

For sinusoidal wave of emf, $K_f = 1.11$

$\therefore E_{rms}$ / phase $= 4.44 K_d K_p \phi f T$ volts ...(3.7)

If the alternator is star-connected, as is usually the case, the line voltage is $\sqrt{3}$ times the phase value.

So line induced emf, $E_L = \sqrt{3} \times 4.44 K_d K_p \phi f T$ volts ...(3.8)

For full-pitched and concentrated windings, $K_p = 1$ and $K_d = 1$